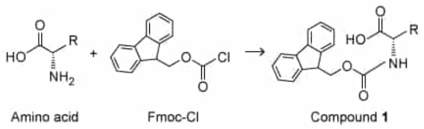
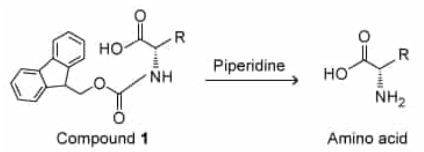
The *N*-terminus of an amino acid must be protected with 9-fluorenylmethoxycarbonyl (Fmoc) during solid state peptide synthesis to avoid side reactions (Scheme I).



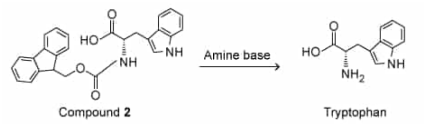
Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

Because the use of piperidine is restricted, structurally similar amines were examined as possible replacements. To determine whether piperidine analogues were suitable alternatives for Fmoc removal, deprotection of Compound **2** was carried out with several amine bases (Scheme III).



Scheme III

Compound **2** was prepared from L-tryptophan and Fmoc-Cl, and the reaction was monitored by thin-layer chromatography (TLC) and visualized by ultraviolet light. The deprotections were done with piperidine and each analogue using a microwave set to 155 watts at 75 °C for 15 seconds. The products of the deprotections were analyzed by high-performance liquid chromatography (HPLC) and mass spectrometry; the retention times of the products were compared to tryptophan and Compound **2** standards.

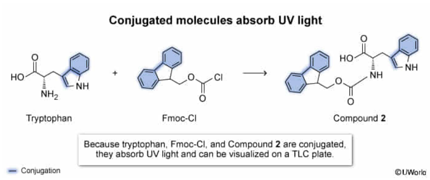
Which of the following statements provides the most plausible explanation for why UV light was used to visualize the thin-layer chromatography plate?

- A. The boiling points of the starting materials and products are far enough apart for separation. [1%]
- B. Tryptophan and Compound **2** will give different molecular ion peaks. [4%]
- C. The amino acid and Fmoc carbonyl bonds stretch and rotate upon UV light absorption. [11%]
- ✔ D. Conjugated double bonds of the aromatic rings on the indole and Fmoc groups absorb UV light. [82%]

Correct

83% answered correctly

Explanation:



Thin-layer chromatography (TLC), a technique that separates components of a mixture based on polarity, can be used to **monitor reactions**. The stationary phase (TLC plate) is made of a polar material, usually silica (SiO_2). Therefore, polar molecules will have a greater affinity for the TLC plate and travel a smaller distance up the plate than nonpolar molecules. After separation of the reaction mixture, the components are visualized. Molecules that have **UV chromophores**, which include double and triple bonds, carbonyls, and conjugated systems, can be visualized with **UV light**.

The aromatic rings of the indole group on the tryptophan side chain and the Fmoc group are made up of conjugated systems of double bonds and therefore can absorb UV light. Because tryptophan, Fmoc-Cl, and compound **2** are visible on a TLC plate under UV light and differ in relative polarity, the addition of the Fmoc protecting group to tryptophan can be monitored by TLC.

(Choice A) Distillation and gas chromatography, not TLC, separate compounds based on their boiling points. They do not use UV light as a means of detection.

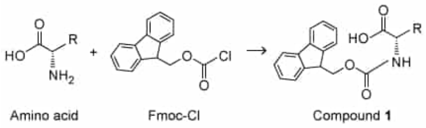
(Choice B) Tryptophan and Compound **2** have different molecular weights and will give different molecular ion peaks, but molecular ion peaks result from ionization in mass spectrometry rather than irradiation with UV light.

(Choice C) Absorption of *infrared* light rather than UV light causes stretching vibrations and rotations of the amino acid and Fmoc carbonyl bonds.

Educational objective:
Thin-layer chromatography, a technique that separates mixture components based on polarity, can be used to monitor reactions. After separation, the mixture components are visualized, usually by UV light. Molecules with UV chromophores (double and triple bonds, carbonyls, conjugated systems) can be visualized with UV light.

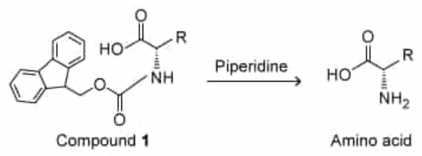
Time spent: 0 sec	**QID**:402392
Subject:	**System:**
Organic Chemistry	Separation Techniques, Spectroscopy, and Analytical Methods

The *N*-terminus of an amino acid must be protected with 9-fluorenylmethoxycarbonyl (Fmoc) during solid state peptide synthesis to avoid side reactions (Scheme I).



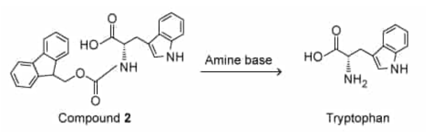
Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

Because the use of piperidine is restricted, structurally similar amines were examined as possible replacements. To determine whether piperidine analogues were suitable alternatives for Fmoc removal, deprotection of Compound **2** was carried out with several amine bases (Scheme III).



Scheme III

Compound **2** was prepared from L-tryptophan and Fmoc-Cl, and the reaction was monitored by thin-layer chromatography (TLC) and visualized by ultraviolet light. The deprotections were done with piperidine and each analogue using a microwave set to 155 watts at 75 °C for 15 seconds. The products of the deprotections were analyzed by high-performance liquid chromatography (HPLC) and mass spectrometry; the retention times of the products were compared to tryptophan and Compound **2** standards.

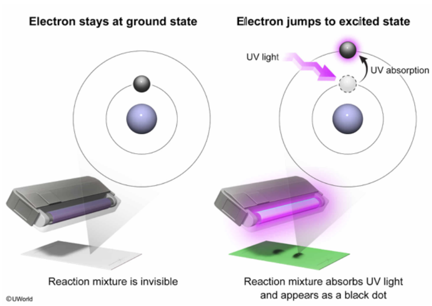
Which of the following most accurately describes the electronic transitions that occur when UV light is used to visualize a sample of the reaction mixture in Scheme III deposited on a thin-layer chromatography plate?

- A. Bound electrons fall to the ground state when absorbing UV light. [2%]
- B. Bound electrons jump to an excited state when absorbing UV light. [89%]
- C. Bound electrons remain at the same energy level when absorbing UV light. [3%]
- D. Bound electrons fall to a lower-energy excited state when absorbing UV light. [4%]

Correct

90% answered correctly

Explanation:



Thin-layer chromatography (TLC) is a technique that separates mixture components based on the polarity of the molecules in the mixture. The mixture components can be visualized under UV light if the molecules contain **UV**

chromophores (ie, portions of a molecule that absorb UV light), including conjugated systems, double or triple bonds, and carbonyls. Absorption of **UV light** by the chromophore **excites electrons** to a **higher energy** state (ie, an **excited state**).

The formation of Compound **2** was monitored by TLC, and the conjugated double bonds in tryptophan, Fmoc-Cl, and Compound **2** each absorb UV light on the TLC plate. The chromophore **electrons jump to a higher energy excited state** as the light is absorbed, and the compounds are visible as black spots on the TLC plate.

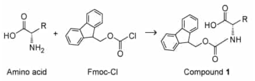
(Choices A and C) Bound electrons do absorb UV light, but absorption of light occurs when electrons are excited to a *higher* energy state rather than falling back to the ground state. If electrons stayed at the same energy level, no light would be absorbed.

(Choice D) If electrons fall from a higher energy level to a lower energy level, light is emitted, not absorbed.

Educational objective:
When UV chromophores (conjugated systems, double and triple bonds, carbonyls) absorb UV light, energy from the UV light excites electrons to a higher energy excited state.

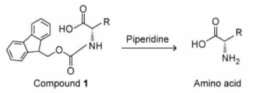
Time spent:0 sec	**QID**:402393
Subject:	**System:**
Organic Chemistry	Separation Techniques, Spectroscopy, and Analytical Methods

The *N*-terminus of an amino acid must be protected with 9-fluorenylmethoxycarbonyl (Fmoc) during solid state peptide synthesis to avoid side reactions (Scheme I).



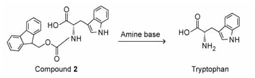
Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

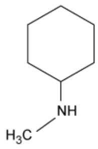
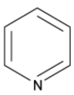
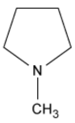
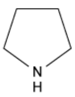
Because the use of piperidine is restricted, structurally similar amines were examined as possible replacements. To determine whether piperidine analogues were suitable alternatives for Fmoc removal, deprotection of Compound **2** was carried out with several amine bases (Scheme III).



Scheme III

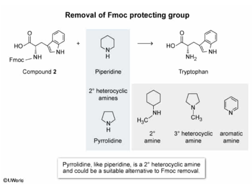
Compound **2** was prepared from L-tryptophan and Fmoc-Cl, and the reaction was monitored by thin-layer chromatography (TLC) and visualized by ultraviolet light. The deprotections were done with piperidine and each analogue using a microwave set to 155 watts at 75 °C for 15 seconds. The products of the deprotections were analyzed by high-performance liquid chromatography (HPLC) and mass spectrometry; the retention times of the products were compared to tryptophan and Compound **2** standards.

Based on information in the passage, which amine was most likely proposed as a replacement for piperidine in Scheme III?

- A.  [18%]
- B.  [20%]
- C.  [5%]
- ✔ D.  [55%]

Correct
55% answered correctly

Explanation:



Amines can be classified as primary, secondary, tertiary, or quaternary (ie, one, two, three, or four nonhydrogen substituents, respectively). The passage states that piperidine is a heterocyclic secondary (2°) amine that is typically used to remove the Fmoc protecting group. Heterocyclic molecules are ring structures that contain at least one heteroatom (neither carbon nor hydrogen) within the ring. Secondary amines are nitrogen atoms bound to two carbon atoms (alkyl and/or aryl groups).

An alternative to piperidine was explored because the use of piperidine is regulated. An amine that is **structurally similar** to piperidine (ie, a heterocyclic 2° amine) would most likely be the best alternative. Pyrrolidine, shown in Choice D, is the only amine given that is both heterocyclic *and* a 2° amine.

(Choice A) This is a secondary amine, but it is not heterocyclic because the nitrogen atom is not part of the ring.

(Choice B) This is an aromatic amine. It contains three carbon-nitrogen bonds, so it is not a secondary amine.

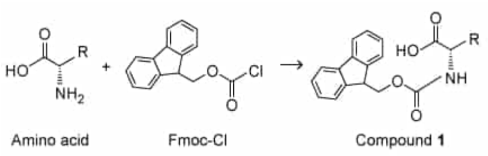
(Choice C) This amine is heterocyclic, but it is a 3° amine rather than a 2° amine.

Educational objective:
Amines can be classified as primary, secondary, tertiary, or quaternary (ie, one, two, three, or four nonhydrogen substituents, respectively).

Time spent: 0 sec
Subject: Organic Chemistry

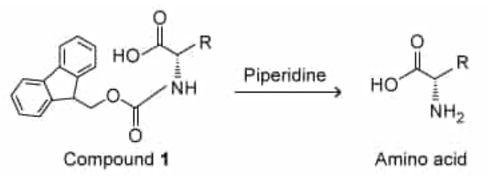
QID: 402394
System: Separation Techniques, Spectroscopy, and Analytical Methods

The *N*-terminus of an amino acid must be protected with 9-fluorenylmethoxycarbonyl (Fmoc) during solid state peptide synthesis to avoid side reactions (Scheme I).



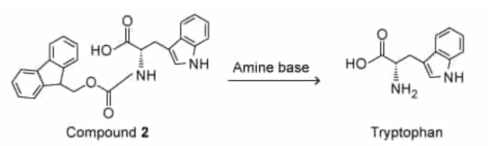
Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

Because the use of piperidine is restricted, structurally similar amines were examined as possible replacements. To determine whether piperidine analogues were suitable alternatives for Fmoc removal, deprotection of Compound **2** was carried out with several amine bases (Scheme III).



Scheme III

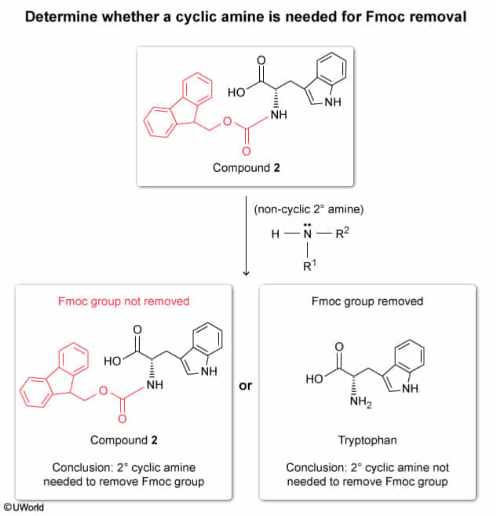
Compound **2** was prepared from L-tryptophan and Fmoc-Cl, and the reaction was monitored by thin-layer chromatography (TLC) and visualized by ultraviolet light. The deprotections were done with piperidine and each analogue using a microwave set to 155 watts at 75 °C for 15 seconds. The products of the deprotections were analyzed by high-performance liquid chromatography (HPLC) and mass spectrometry; the retention times of the products were compared to tryptophan and Compound **2** standards.

Which reaction would best give data to support the need of a cyclic amine for removal of the Fmoc group?

- A. React tryptophan with a secondary noncyclic amine and determine whether the Fmoc group was removed. [5%]
- B. React tryptophan with a tertiary noncyclic amine and determine whether the Fmoc group was removed. [2%]
- ✔ C. React Compound **2** with a secondary noncyclic amine and determine whether the Fmoc group was removed. [87%]
- D. React Compound **2** with a tertiary noncyclic amine and determine whether the Fmoc group was removed. [5%]

Correct
87% answered correctly

Explanation:



Conclusions about a **variable** can be **drawn** from **experimental data**. To determine whether a result is dependent on a specific variable, the experiment is replicated and the variable in question is changed while all other variables are held constant so that only **one parameter** of the experiment is **changed**. Experimental setups in which more than one variable is changed cannot be compared to the original experiment because differences in the results cannot be attributed to any one variable.

The passage states that piperidine, a heterocyclic secondary amine, is used to remove Fmoc, a protecting group. The two variables in the amine's structure are the substitution of the nitrogen atom (secondary) and whether the amine is cyclic. To conclude that the amine must be cyclic for Fmoc group removal, a noncyclic amine should be used for comparison.

Therefore, researchers should react Compound **2** with a noncyclic secondary amine and determine whether the Fmoc group was removed. If the Fmoc group is removed by a noncyclic amine, the amine does *not* need to be cyclic for deprotection, whereas if the Fmoc group is not removed, the cyclic nature of the amine is important for deprotection.

(Choices A and B) Tryptophan does not contain an Fmoc group. Therefore, treating it with an amine would not contribute any data to support the conclusion that a cyclic amine is necessary for Fmoc removal.

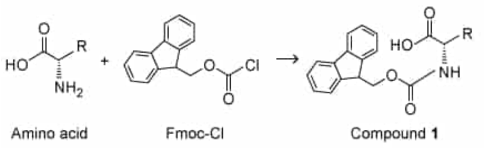
(Choice D) Using a tertiary noncyclic amine to attempt Fmoc group removal would not help support the conclusion that amines must be cyclic to remove Fmoc because both variables (cyclic nature and substitution) were changed. If Fmoc removal does not occur, it would be unclear whether it was because the amine is noncyclic or because the amine is tertiary, or both.

Educational objective:
Conclusions about a variable can be drawn from experimental data. To determine whether a result is dependent on a specific variable, the experiment is repeated and the variable in question is changed while all other variables are held constant so that only one parameter of the experiment changes.

Time spent:0 sec
Subject:
Organic Chemistry

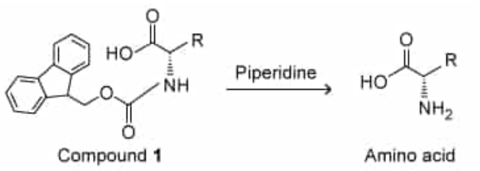
QID:402395
System:
Separation Techniques, Spectroscopy, and Analytical Methods

The *N*-terminus of an amino acid must be protected with 9-fluorenylmethoxycarbonyl (Fmoc) during solid state peptide synthesis to avoid side reactions (Scheme I).



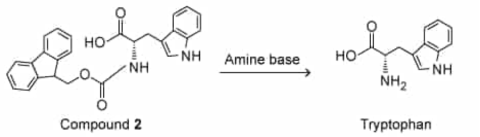
Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

Because the use of piperidine is restricted, structurally similar amines were examined as possible replacements. To determine whether piperidine analogues were suitable alternatives for Fmoc removal, deprotection of Compound **2** was carried out with several amine bases (Scheme III).



Scheme III

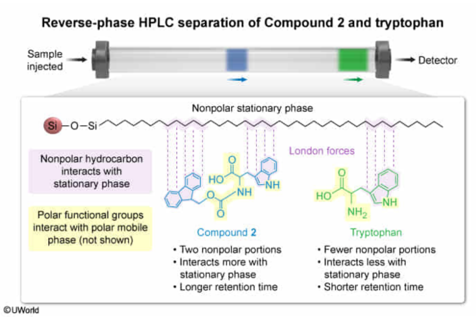
Compound **2** was prepared from L-tryptophan and Fmoc-Cl, and the reaction was monitored by thin-layer chromatography (TLC) and visualized by ultraviolet light. The deprotections were done with piperidine and each analogue using a microwave set to 155 watts at 75 °C for 15 seconds. The products of the deprotections were analyzed by high-performance liquid chromatography (HPLC) and mass spectrometry; the retention times of the products were compared to tryptophan and Compound **2** standards.

Considering the difference in the number of hydrocarbon bonds in the structures of tryptophan and Compound **2**, what can be inferred about the HPLC column used in the passage if tryptophan has a shorter retention time than Compound **2**?

- ✔ A. A reverse-phase HPLC column was used with a nonpolar stationary phase and polar mobile phase. [44%]
- B. A reverse-phase HPLC column was used with a polar stationary phase and nonpolar mobile phase. [8%]
- C. A normal-phase HPLC column was used with a polar stationary phase and nonpolar mobile phase. [33%]
- D. A normal-phase HPLC column was used with a nonpolar stationary phase and polar mobile phase. [13%]

Correct
44% answered correctly

Explanation:



High-performance liquid chromatography (HPLC) is a purification technique used for small sample sizes. The instrumentation consists of a sample injector, a column (stationary phase), a solvent under pressure (mobile phase), a detector, and a computer for data acquisition. **Two types** of **columns**—normal phase (NP) or reverse phase (RP)—can be used, depending on the relative polarity of the compounds being separated.

RP-HPLC uses a **nonpolar stationary phase** (typically a C-18 alkyl hydrocarbon) and a comparatively **polar mobile phase**. In RP-HPLC, the least polar compound in a mixture will have the strongest interaction with the nonpolar stationary phase and therefore the longest retention time.

Compared to tryptophan, Compound **2** contains two additional functional groups (ie, a polycyclic aromatic hydrocarbon and an ester) that are part of the Fmoc protecting group. While the ester adds 2 polar C–O bonds to Compound **2**, the polycyclic aromatic hydrocarbon adds 27 nonpolar bonds (ie, 11 C–H bonds and 16 C–C bonds), which significantly increases the nonpolar, hydrophobic character of Compound **2**. Consequently, Compound **2** is *more nonpolar* than tryptophan. The question states that when analyzed by HPLC as described in the passage, tryptophan has a shorter retention time than Compound **2**. Therefore, tryptophan interacts more with the mobile phase, and Compound **2** has more interactions (ie, more London forces) with the stationary phase. Because tryptophan has fewer nonpolar portions (ie, is more polar) than Compound **2**, the mobile phase must be more polar than the stationary phase. A relatively polar mobile phase and nonpolar stationary phase indicate that an RP-HPLC column was used.

(Choice B) An RP-HPLC column was used, but RP columns have a *nonpolar* stationary phase and a *polar* mobile phase.

(Choices C and D) In NP-HPLC, the mobile phase is nonpolar compared to the stationary phase; therefore, compounds with fewer polar bonds have a greater affinity for the mobile phase and travel through the column faster than more polar compounds. Tryptophan would have a *longer* retention time than Compound **2** on an NP-HPLC column.

Educational objective:

In high-performance liquid chromatography (HPLC), two types of columns—normal phase (NP) or reverse phase (RP)—can be used, depending on the polarity of the compounds being separated. NP-HPLC uses a polar stationary phase and a nonpolar mobile phase, whereas RP-HPLC uses a nonpolar stationary phase and a polar mobile phase.

Time spent:0 sec
Subject:
Organic Chemistry

QID:402396
System:
Separation Techniques, Spectroscopy, and Analytical Methods

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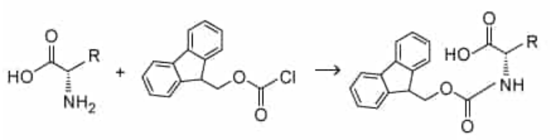
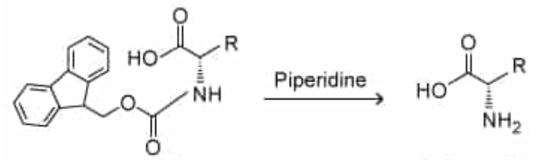


Scheme I

The Fmoc protecting group is typically removed with piperidine (Scheme II), a heterocyclic secondary amine.



Scheme II

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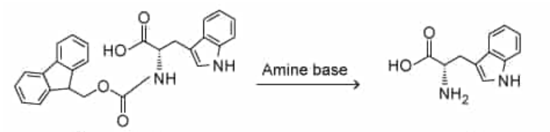

Scheme III

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If a C18 hydrocarbon HPLC column is used for analysis of the Scheme III reaction results, which of the following solvents would NOT be used for the mobile phase?

- A. Acetone [2%]
- B. Methanol [4%]
- ✔ C. Hexanes [81%]
- D. Water [11%]

Correct
80% answered correctly

Explanation:

Normal-phase HPLC	Reverse-phase HPLC
• Polar stationary phase *Silica (SiO_2)* • Nonpolar mobile phase *Hexanes*	• Nonpolar-stationary phase *C18 Hydrocarbon* • Polar mobile phase *Water, methanol, acetone*

High-performance liquid chromatography (HPLC) is a technique that can be used to analyze compound purity, separating components of a mixture based on polarity. There are two types of stationary and mobile phases that are used (polar or nonpolar), depending on the polarity of the compounds being separated. A **polar stationary phase** (usually silica SiO_2) is used in conjunction with a **nonpolar mobile phase**, and this is known as normal-phase HPLC (NP-HPLC). Reverse-phase HPLC (RP-HPLC) consists of a **nonpolar stationary phase**, such as C18 hydrocarbon, and a **polar mobile phase**.

Solvents can be classified as polar or nonpolar. Polar solvents contain electronegative atoms, such as oxygen, that create a dipole. Nonpolar solvents have less electronegative atoms and a temporary dipole, which causes London forces.

The use of a C18 hydrocarbon HPLC column indicates that the stationary phase is nonpolar, and therefore corresponds to RP-HPLC. A polar solvent must be used as the mobile phase in RP-HPLC. Acetone, methanol, and water are all polar solvents **(Choices A, B, and D)**. These solvents contain oxygen (electronegative), have dipoles, and experience dipole-dipole interactions with one another.

Hexanes is a nonpolar solvent and would be used as a mobile phase in NP-HPLC but would *not* be used in RP-HPLC.

Educational objective:
High-performance liquid chromatography (HPLC) is a technique that separates compounds based on polarity. The two types of HPLC are normal phase (NP) and reverse phase (RP), in which NP-HPLC uses a polar column (stationary phase) and a nonpolar solvent (mobile phase) and RP-HPLC uses a nonpolar column and a polar solvent.

Time spent: 0 sec	**QID**:402397
Subject: Organic Chemistry	**System:** Separation Techniques, Spectroscopy, and Analytical Methods